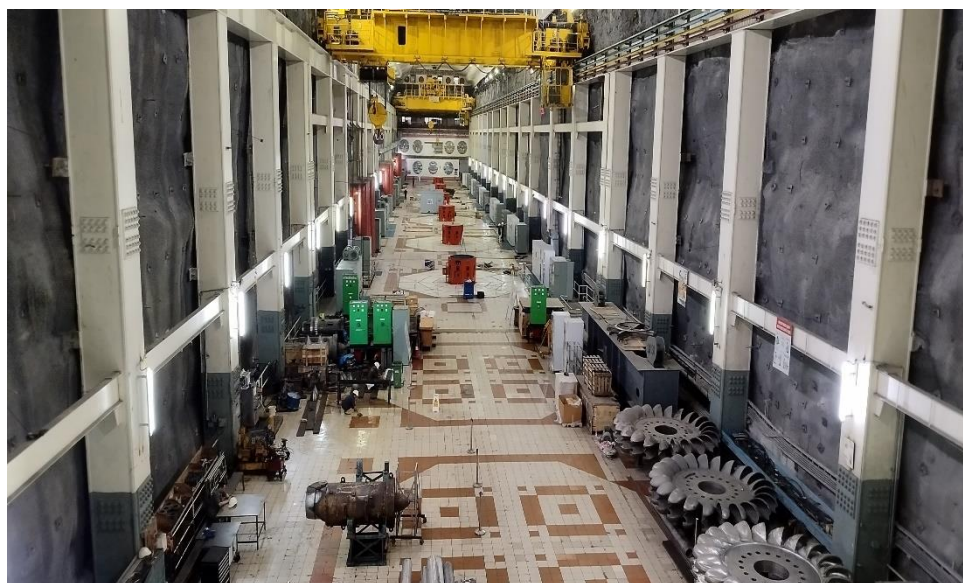




Annual Maintenance Completion Report (2025-2026)

Generating unit VI



Mechanical Maintenance Unit

Tala Hydropower Plant

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1. Overview

The Annual Maintenance of Unit 6 was carried out after unit 2, focusing on key inspection and servicing activities to ensure the continued reliability of the generating unit. While the major works included replacement of MIV seals, other works such as pre-testing and overhauling of nozzles, inspection of the runner, replenishment of nitrogen in the MIV accumulator bank, calibration of MIV pressure gauges, cleaning of the Main Heat Exchanger, and inspection of the Cooling Water pumps were carried out. Additionally, Ultrasonic Testing (UT) was performed on the MIV decompression pipeline, bypass pipeline, and the pressure accumulators of both the MIV and governor systems. All activities were completed in close coordination with relevant sections, maintaining compliance with safety and quality standards.

Scheduled vs Actual date of start and completion of maintenance on the generating unit is shown in Table-1

SN	Unit #	Approved Schedule			Actual Completion		
		Start	End	No of Days	Start	End	No of Days
1.	Unit VI	25.04.2026	31.05.2026	31	01.012.2025	24.02.2026	67

Table 1: Unit 6 AM schedule

The Annual Maintenance work of the unit commenced on 01.12.2025. Although the maintenance was originally scheduled from 25.04.2026, the work was advanced due to the urgent requirement for replacement of the MIV seals, as heavy leakage was observed across the MIV. Replacement of both seals of MIV was carried out along with the regular maintenance activities because of which the AM period was extended till 17th February, 2026. This delayed the spinning of the unit after completion of AM. The unit was put to mechanical spin at 07:45 hours on 25th February, dry run was initiated and shut down upon reaching the temperature limit of the bearings. The machine was successfully at 14:01 hours on 27th February.

The major maintenance activities conducted for the unit are detailed hereunder and all test/calibration reports, checklists and inspection notes are placed as Annexure.

2. Turbine

Table 2: Spares consumed for Turbine system

S. No	Materials	Quantity
1	M10x60mm long Half-threaded Hexa bolt	40
2	H.T Allen bolt, M20x150 H/thread, Gr.12.9	5
3	PTFE ring seal 200X209.2X7.5mm with inner,	1
4	Wiper seal PU with casing 110X120X7/10mm,	2
5	NBR Endless O-ring 320x6mm 70 shore	10
6	NBR Endless O-ring 335x6mm 70 shore	2

7	Cup Seal P.P-110X130X12mm	2
8	NBR Endless O-ring 19.5x3mm 70 shore	2
9	Cup seal P.P-200x230x16mm	2
10	NBR Endless O-ring 246x6mm 70 shore	1
11	Piston seal, 220x199x7.7mm (OD*ID*thickness)	6
12	H.T Allen bolt, M16x160, grade12.9,	1
13	SS Allen bolt, M12x60, Gr9.8, half thread,	12
14	SS Allen bolt, M12x80, Gr.9.8, half thread,	22
15	Locking nut KMTA 18 for nozzle	1
16	NBR Endless O-ring 28x3mm 70 shore	2
17	Wiper seal PU with casing200X220X12/16mm,	2
18	MS Plain Washer, M8	60
19	SS Allen bolt, M36x340, grade 9.8,	4
20	SS Allen bolt, M20x200, grade 9.8,	1
21	Argon gas 7.00M3/cylinder	21
22	Multipurpose grease, 3	2

2.1. Nozzle Injector Assembly

All five nozzle injector assemblies underwent functional verification through in-situ oil pressure testing, conducted at an initial pressure of 100 bar for 30 minutes. No signs of failure were observed in any of the nozzles. However, nozzle 2 was dismantled for overhaul as part of the Annual Maintenance, having completed over seven years of operation without servicing. For the remaining assemblies that were not dismantled, the needle tips and liner assemblies were inspected, and fasteners were replaced as required.



Figure 1: Overhauling & servicing of nozzle injectors

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2.2 Runner

Table 3: Runner particulars

Particulars	Runner (1020)
Make	Andritz Hydro, Switzerland
Type	Forged Fabricated and hard coated
Hard Coating	HVOF
Cumulative Running Hours	59,187.36 hours as of 01.12.2025
Repair/ reclamation History	Reclaimed thrice at BHSL; between November 2016 and June 2018 after 34,079 running hours, between October 2018 and December 2018 after 2377 hours, and between January 2020 and March 2022 after 4535.29 hours

The unit VI runner 1020 was in operation from 2011 and its running hours since the commission has reached 59,187.36 hours by the time the unit was taken under shutdown for annual maintenance on 01.12.2025.



Figure 2: Erosion observed on runner buckets

The NDT personnel from HR&DC examined the physical condition of all the runner buckets after visual inspection with Magnetic particle testing on the roots and eroded parts on all 22 buckets. Following the dam overtopping incident on 5th October, the runner buckets were found to have some deep erosions which needed weld repair. The in-situ weld repair was carried out.

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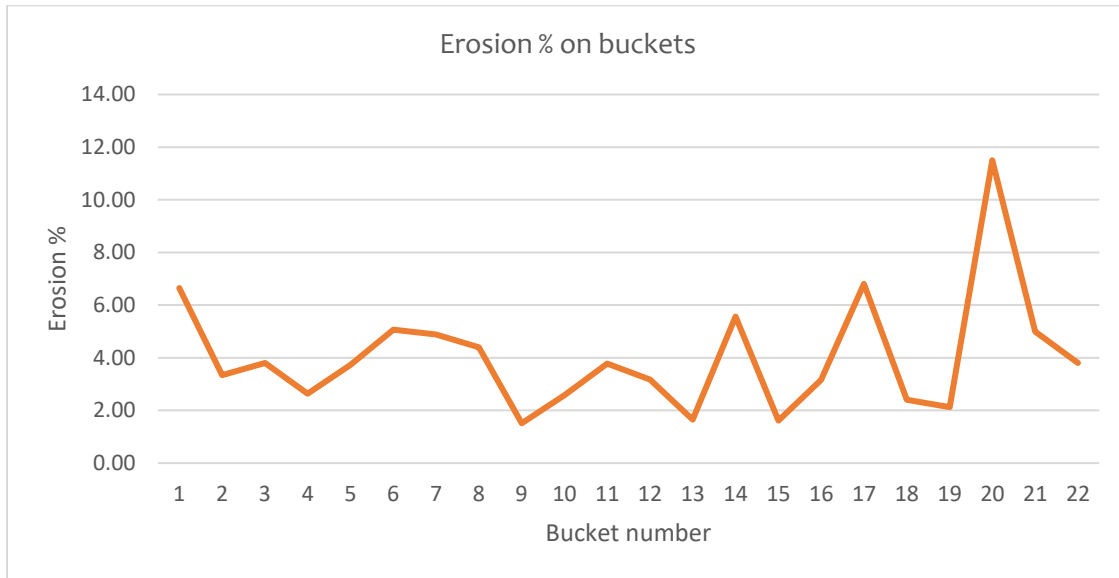


Figure 3: Erosion rates

The above chart gives information on parent material erosion on each bucket of the runner. The erosion is recorded maximum of 11.5% on bucket 20 while the least is observed on bucket 11 whose erosion is about 1.5%

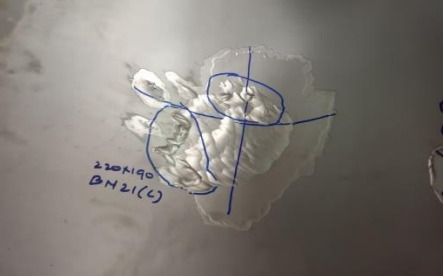
The weld repair was performed in parallel with the nozzle servicing activities. Upon completion, the bucket surfaces were ground to achieve a smooth finish, removing surface irregularities and enhancing flow characteristics. Magnetic Particle Testing was then carried out to verify that no defects remained.

Weld repair using GTAW was carried out on buckets 20, 21, 7 and others, as shown in the pictures below.

Table 4: Weld repair on buckets

Bucket number	Before repair	After repair
20 (R)		
20 (L)		

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7 (L)		
21 (L)		

3. Governor and MIV System

Table 5: Spares consumed for Governor and MIV system

S. No	Description	Quantity
1	Filter element-0160 D 020 BN4HC	1
2	Filter element-0160 DN 010 BN/HC	1
3	Governor OPU return line duplex filter,	2

3.1 Governing System

The main function of governor system is to maintain the rotational speed of turbine within acceptable limits during no-load condition. It performs this function by sensing changes in speed and accordingly controlling flow of water into turbine at fast rate to maintain the speed with reference to grid frequency.

For this unit, the SCADA implementation had been successfully completed during the previous Annual Maintenance. Accordingly, no works related to the HMC or NCC were required this year, unlike in past maintenance cycles where such activities were a major part of the scope. Instead, routine servicing tasks were carried out to ensure system reliability. These included thorough cleaning of the OPU tank, inspection of pump and oil filters, and replenishment of nitrogen pressure in the accumulator bank to maintain optimal system performance. All high-pressure pipelines associated with VTA (Valve Table Assembly) were carefully inspected for any signs of leakage, and necessary tightening or corrective measures were taken wherever required.

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3.1.1 Valve Table Assembly (VTA)



Figure 4: Valve Table Assembly

The VTA is a critical component of the hydraulic control system, primarily responsible for directing and regulating oil flow to the servomotors of the needle and deflector systems. It houses the solenoid valves, pressure gauges, and other control elements that interface with the SCADA system for remote operation and monitoring.

With the SCADA integration completed during the previous Annual Maintenance, the VTA now facilitates more precise and efficient control of the turbine's hydro-mechanical functions, reducing the need for manual intervention and improving overall system responsiveness.

Deflector gap setting was done and the set result is tabulated below.

Table 6: Deflector gap setting readings

Nozzle	Initial reading (mm)	Final reading (mm)
J1	9	9
J2	13	8
J3	9	8
J4	11	8
J5	7	8

3.2 MIV System

The MIV system is the primary isolation for turbine and principal protection for the station against penstock water. The major work activity this maintenance in the unit was the replacement of MIV seals. Heavy leakage had been observed from both the seals during the peak months. Particularly after the dam overtopping incident on October, replacement of both the seals had been planned.



Figure 5: MIV seal replacement

For the replacement of both maintenance and service seals, the entire MIV body was lifted to the service bay. Following overhauling and detailed inspection, the door seating rings and body sealing rings were replaced, after which leakage testing was carried out. The total duration of the work was 41 days.

Signature

A comprehensive report on the MIV seal replacement will be provided in a separate submission.

During the MIV seal replacement, the internal surfaces of the distributor, particularly at the pipe joint sections, were scraped and repainted. These areas are subjected to continuous water pressure and are therefore prone to wear over time. The distributor directs high-pressure water from the MIV into high-velocity jets aimed precisely at the buckets of the runner through 5 nozzles.



Figure 6: Scraping and painting in distributor



Figure 7: Erosion on MIV bypass pipeline

The internal sections of the bypass pipeline were found to be eroded and were subsequently repaired by welding. As this section is made of MS and operates under high pressure, leaving the erosion unattended could pose a significant risk during operation.

The pressure gauges of MIV control panel and accumulator have been calibrated at HRDC and four gauges had pressure deviations beyond acceptable ranges, which will be replaced after the receipt of new gauges. The report is attached.

In order to ensure safe parameters of the control pipelines for safe and smooth operations, periodic NDT is highly recommended to be done annually since these are considered among the most critical parts.

The ultrasonic testing of MIV bypass pipeline, decompression pipeline and pressure accumulators for thickness verification was done by an inspection team from HRDC to inspect any possible deviations in the pipeline thickness or any crack development before it amplifies. The test result has been attached.

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Figure 6: UT on high-pressure pipelines

The servo valve of the decompression pipeline was opened and inspected for debris. This activity is recommended during every annual maintenance and also during runner inspections to ensure that no foreign particles or sediments obstruct the valve operation, which could affect the proper functioning of the decompression system and lead to abnormal pressure conditions.

OPU pump capacity was checked and found normal. Functionality check for internal leakages and servicing was carried on all the directional control solenoid valves provided in OPU and MIV control panel. Contactors and limit switches of MIV were checked for their healthiness and replacement done wherever necessary. During the functionality check, no N₂ gas leakage was observed.

4. Cooling Water System

4.1. Clean Cooling Water (CCW) System

During this Annual Maintenance, both pumps underwent a visual inspection. No overhauling or replacement of parts was carried out.

Following the inspection, shaft alignment between the pump and motor was verified using an Easy-Laser alignment system. The laser alignment process confirmed that both angular and parallel alignments were within acceptable tolerance limits, minimizing potential vibration and mechanical stress on the coupling.



Figure 7: Overhauling and servicing of CCW pump

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4.2. Raw Cooling Water (RCW) System

The pumps were visually inspected, and no overhauling was performed. During the inspection, the bearing hubs and pump casings were cleaned to remove accumulated debris and contaminants. The flowmeter of the RCW outlet was replaced, along with the flowmeters for the MIV and TGB cooling lines. The valve handle of RCW pump 1 was also replaced.

Additionally, the motorized strainer and cyclone separator were thoroughly cleaned to ensure smooth filtration and uninterrupted flow. These preventive maintenance activities were conducted to help maintain overall system performance and reliability.

4.3. Heat Exchanger

Each unit is equipped with a shell-and-tube type heat exchanger (MHE) to facilitate the circulation of clean cooling water for the stator air coolers, OPU coolers, and bearing coolers. Raw cooling water is supplied through the tubes inside the heat exchanger to enable heat transfer.

During the annual maintenance, the heat exchanger was opened, and the tubes were thoroughly cleaned and flushed to remove any scaling or deposits that could impede heat transfer efficiency. The flowmeter at the outlet was replaced to ensure accurate monitoring of the flow. After cleaning, the end covers were reassembled, and pressure testing was conducted to ensure there were no leaks and that the unit was fit for continued operation.



Figure 8: Cleaning of Main Heat Exchanger

5. Internal/ external services availed

- Runner inspection (Magnetic Particle Inspection) by HRDC
- Ultrasonic testing on decompression, bypass pipelines, pressure accumulators and distributor pipeline joints by HRDC
- Vibration analysis of CCW and RCW pumps by HRDC
- Calibration of pressure gauges on MIV panels by HRDC
- Testing of Governor and MIV system oil samples by HRDC



6. Maintenance Challenges

- One major challenge was the re-fitment of the MIV seals due to dimensional mismatch during the seal replacement. The seals had to be returned to BHSL for corrective re-machining to achieve proper fitment, which consequently extended the maintenance duration.
- Repair of the severely eroded interior section of the MIV bypass outlet pipe was also a challenge. The location of the damaged area had poor accessibility, particularly for grinding, which restricted proper surface preparation and made the repair work difficult.

7. Any Modification and improvement

- The jockey pump installed earlier in the MIV OPU system was commissioned during this Annual Maintenance by CPU team. The jockey pump is provided to maintain and retain the required oil pressure in the OPU system.



Annexure

S.NO	Annexure	Description
1	Annexure 1	Auxiliary Final checklist
2	Annexure 2	Deflector Mechanical Gap setting
3	Annexure 3	Deflector Servo timing set
4	Annexure 4	Fasteners Tightness checklist
5	Annexure 5	G100 Governor system Functionality Checklist
6	Annexure 6	Governor Final checklist
7	Annexure 7	Governor Nitrogen Pressure retention checklist
8	Annexure 8	Governor OPU Safety valve setting
9	Annexure 9	Governor OPU Pump Capacity Checklist
10	Annexure 10	Governor Pre-checklist
11	Annexure 11	Governor system pressure retention Checklist
12	Annexure 12	Inspection of Decompression servo valve
13	Annexure 13	MIV Final checklist
14	Annexure 14	MIV OPU Safety valve setting
15	Annexure 15	MIV pre-checklist
16	Annexure 16	MIV Pressure Retention Checklist
17	Annexure 17	MIV system timing set
18	Annexure 18	Nozzle Assembly checklist
19	Annexure 19	Nozzle post assembly checklist
20	Annexure 20	Nozzle pre-checks
21	Annexure 21	Nozzle timing check
22	Annexure 22	Turbine Final checklist
23	Annexure 23	Underwater components Inspection (Before weld repair)
24	Annexure 24	Underwater components Inspection (After weld repair)
25	Annexure 25	Magnetic Particle Testing report for runner
26	Annexure 26	Pressure Gauge Calibration Report
27	Annexure 27	UT report on pressure accumulators
28	Annexure 28	UT report on MIV high-pressure pipelines
29	Annexure 29	Pump Vibration assessment report

Submitted by:


 (Sharmila Rai)

Assistant Engineer

Mechanical Maintenance Unit

Tala Hydropower Plant